File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - utils.py - PyCharm

HelloSpark 〉 lib 〉 utils.py

HelloSpark

Project

- HelloSpark C:\Users\prashant\Documents\GitHub\
  - app-logs
  - data
    - sample.csv
  - lib
    - __init__.py
    - logger.py
    - utils.py
  - HelloSpark.py
  - log4j.properties
  - spark.conf
- External Libraries
- Scratches and Consoles

HelloSpark.py utils.py

```python

from pyspark import SparkConf


def get_spark_app_config():
    spark_conf = SparkConf()
    config = configparser.ConfigParser()
    config.read("spark.conf")

    for (key, val) in config.items("SPARK_APP_CONFIGS"):
        spark_conf.set(key, val)
    return spark_conf


def load_survey_df(spark, data_file):
    return spark.read \
        .option("header", "true") \
        .option("inferSchema", "true") \
        .csv(data_file)


def count_by_country(survey_df):
    return survey_df \
        .where("Age < 40") \
        .select("Age", "Gender", "Country", "state") \
        .groupBy("Country") \
        .count()
```

Formatted 1 line

File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - spark.conf - PyCharm

HelloSpark > spark.conf

HelloSpark

Project

HelloSpark C:\Users\prashant\Documents\GitHub\

- app-logs
- data
  - sample.csv
- lib
  - __init__.py
  - logger.py
  - utils.py
- HelloSpark.py
- log4j.properties
- spark.conf

External Libraries

Scratches and Consoles

HelloSpark.py | spark.conf | sample.csv | utils.py

```
[SPARK_APP_CONFIGS]
spark.app.name = Hello Spark
spark.master = local[3]
spark.sql.shuffle.partitions = 2
```

File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - HelloSpark.py - PyCharm

HelloSpark > HelloSpark.py

Project

- HelloSpark C:\Users\prashant\Documents\GitHub\
  - app-logs
  - data
    - sample.csv
  - lib
    - __init__.py
    - logger.py
    - utils.py
  - HelloSpark.py
  - log4j.properties
  - spark.conf
- External Libraries
- Scratches and Consoles

HelloSpark.py | spark.conf | sample.csv | utils.py

```
    logger = Log4J(spark)

    if len(sys.argv) != 2:
        logger.error("Usage: HelloSpark <filename>")
        sys.exit(-1)

    logger.info("Starting HelloSpark")

    survey_df = load_survey_df(spark, sys.argv[1])
    partitioned_survey_df = survey_df.repartition(2)

    count_df = count_by_country(partitioned_survey_df)

    logger.info(count_df.collect())

    input("Press Enter")
    logger.info("Finished HelloSpark")
    # spark.stop()
if __name__ == "__main__"
```

Run: HelloSpark

```
C:\Users\prashant\anaconda3\envs\HelloSpark\python.exe C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py data\sample.csv
20/06/09 10:29:50 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

Git | TODO | Run | Terminal | Python Console | Event Log

28:1 CRLF UTF-8 4 spaces Python 3.7 (HelloSpark) master

Hello Spark - Spark Jobs

localhost:4040/jobs/

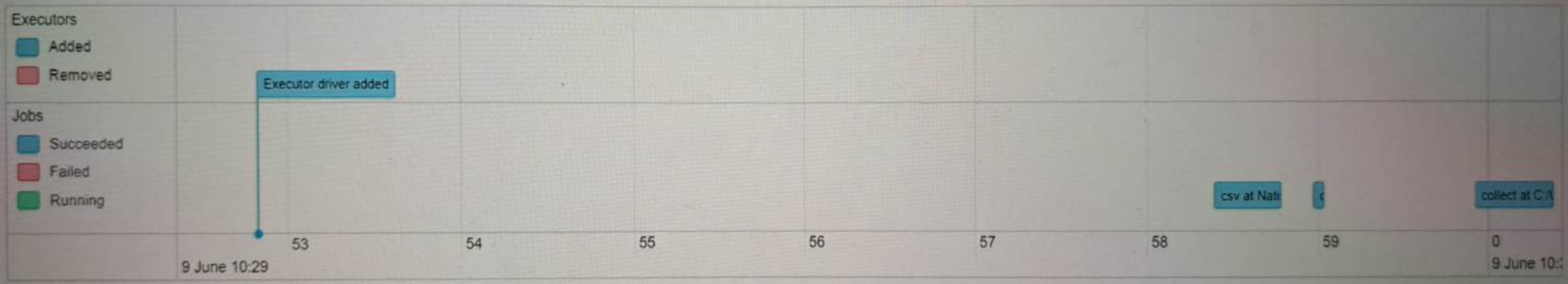
spark 2.4.5

Spark Jobs (?)

User: prashant
Total Uptime: 33 s
Scheduling Mode: FIFO
Completed Jobs: 3

Event Timeline

Enable zooming



Completed Jobs (3)

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
2	collect at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.p... collect at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py:27	2020/06/09 10:29:59	0.5 s	3/3	5/5
1	csv at NativeMethodAccessorImpl.java:0 csv at NativeMethodAccessorImpl.java:0	2020/06/09 10:29:58	63 ms	1/1	1/1
0	csv at NativeMethodAccessorImpl.java:0 csv at NativeMethodAccessorImpl.java:0	2020/06/09 10:29:58	0.4 s	1/1	1/1

Hello Spark - Stages for All Jobs

localhost:4040/stages/

Spark 2.4.5 | Jobs | Stages | Storage | Environment | Executors | SQL | Hello Spark application UI

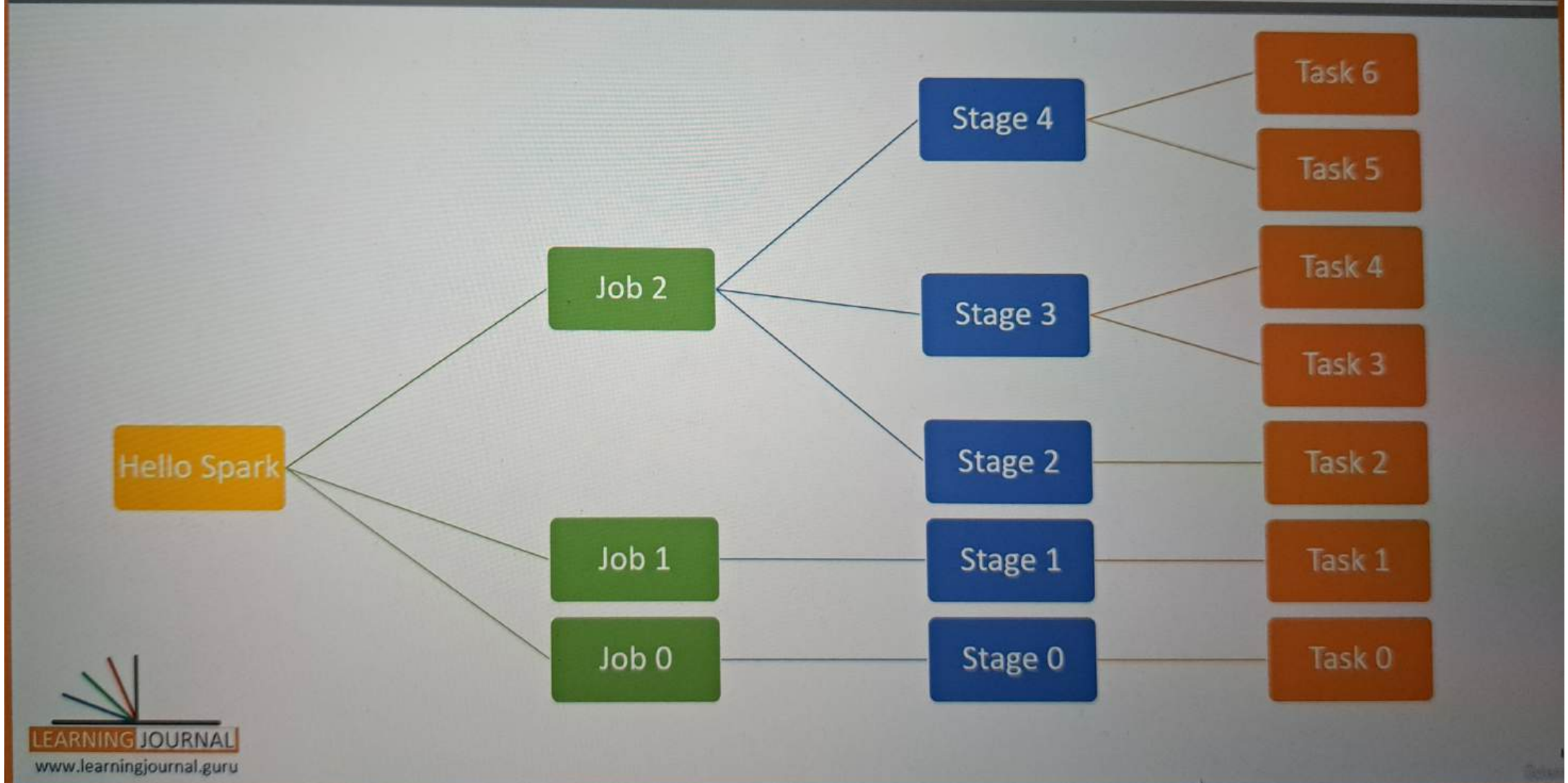
Stages for All Jobs

Completed Stages: 5

Completed Stages (5)

Stage Id	Description		Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
4	collect at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py:27	+details	2020/06/09 10:30:00	15 ms	2/2			229.0 B	
3	collect at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py:27	+details	2020/06/09 10:30:00	0.2 s	2/2			232.0 B	229.0 B
2	collect at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py:27	+details	2020/06/09 10:29:59	0.2 s	1/1	2.2 KB			232.0 B
1	csv at NativeMethodAccessorImpl.java:0	+details	2020/06/09 10:29:58	47 ms	1/1	2.2 KB			
0	csv at NativeMethodAccessorImpl.java:0	+details	2020/06/09 10:29:58	0.3 s	1/1	2.2 KB			

Spark Execution Plan

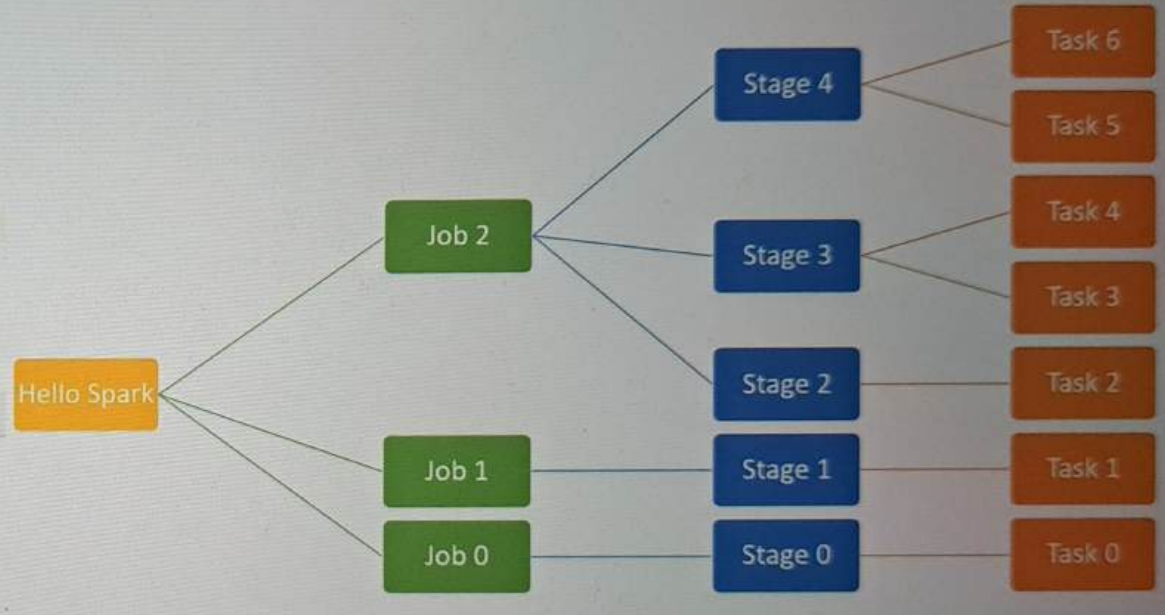


Spark Execution Plan

```python
    logger.info("Starting HelloSpark")

    survey_df = load_survey_df(spark, sys.argv[1])
    partitioned_survey_df = survey_df.repartition(2)

    count_df = count_by_country(partitioned_survey_df)

    logger.info(count_df.collect())

    input("Press Enter")
    logger.info("Finished HelloSpark")
    # spark.stop()
```

utils.py

```python
def count_by_country(survey_df):
    return survey_df \
        .where("Age < 40") \
        .select("Age", "Gender", "Country", "state") \
        .groupBy("Country") \
        .count()
```



LEARNING JOURNAL
www.learningjournal.guru

HelloSpark.py × utils.py ×

```python
from pyspark import SparkConf


def get_spark_app_config():
    spark_conf = SparkConf()
    config = configparser.ConfigParser()
    config.read("spark.conf")

    for (key, val) in config.items("SPARK_APP_CONFIGS"):
        spark_conf.set(key, val)
    return spark_conf


def load_survey_df(spark, data_file):
    return spark.read \
        .csv(data_file)
    # .option("header", "true") \
    # .option("inferSchema", "true") \


def count_by_country(survey_df):
    return survey_df \
        .where("Age < 40") \
        .select("Age", "Gender", "Country", "state") \
        .groupBy("Country") \
```

HelloSpark.py × utils.py ×

```python
import ...

if __name__ == "__main__":
    conf = get_spark_app_config()
    spark = SparkSession.builder \
        .config(conf=conf) \
        .getOrCreate()

    logger = Log4J(spark)

    if len(sys.argv) != 2:...

    logger.info("Starting HelloSpark")

    survey_df = load_survey_df(spark, sys.argv[1])
    # partitioned_survey_df = survey_df.repartition(2)

    # count_df = count_by_country(partitioned_survey_df)

    # logger.info(count_df.collect())

    input("Press Enter")
    logger.info("Finished HelloSpark")
    # spark.stop()
```

Hello Spark - Spark Jobs

localhost:4040/jobs/

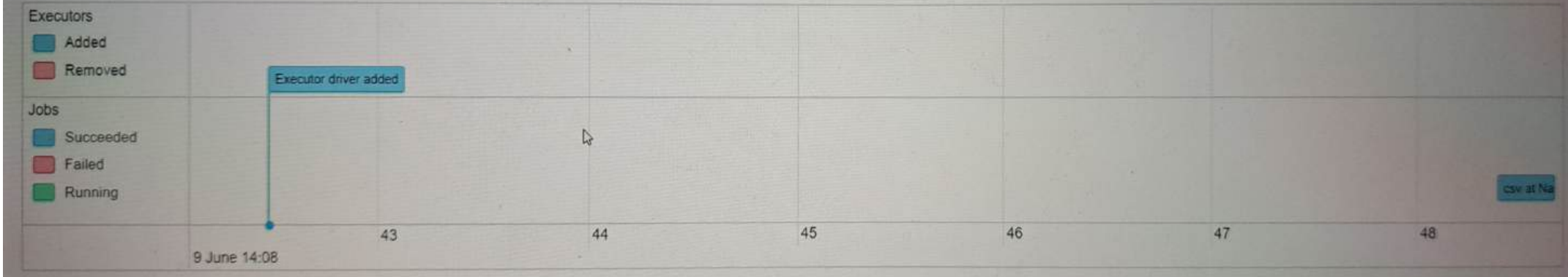
Spark 2.4.5 | Jobs | Stages | Storage | Environment | Executors | SQL | Hello Spark application UI

Spark Jobs (?)

User: prashant
Total Uptime: 19 s
Scheduling Mode: FIFO
Completed Jobs: 1

Event Timeline

Enable zooming



Completed Jobs (1)

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
0	csv at NativeMethodAccessorImpl.java:0 csv at NativeMethodAccessorImpl.java:0	2020/06/09 14:08:48	0.3 s	1/1	1/1

Hello Spark - Stages for All Jobs

localhost:4040/stages/

Spark 2.4.5 | Jobs | Stages | Storage | Environment | Executors | SQL | Hello Spark application UI

Stages for All Jobs

Completed Stages: 1

Completed Stages (1)

Stage Id	Description		Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
0	csv at NativeMethodAccessorImpl.java:0	+details	2020/06/09 14:08:48	0.2 s	1/1	2.2 KB			

Hello Spark - Details for Stage 0

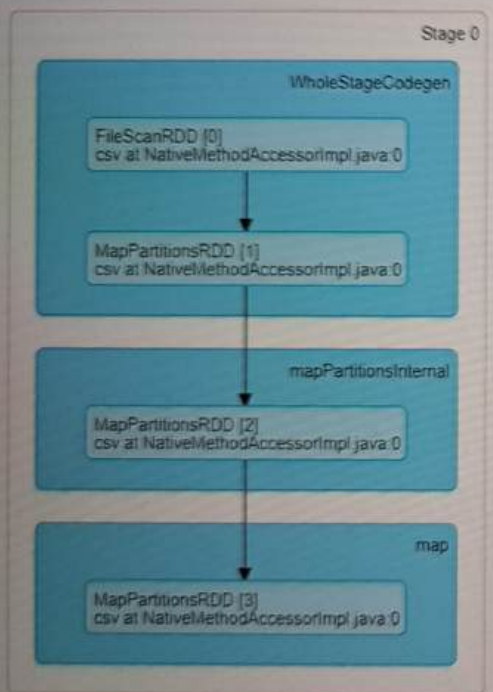
localhost:4040/stages/stage/?id=0&attempt=0

Spark 2.4.5 | Jobs | Stages | Storage | Environment | Executors | SQL | Hello Spark application UI

Details for Stage 0 (Attempt 0)

Total Time Across All Tasks: 0.1 s
Locality Level Summary: Process local: 1
Input Size / Records: 2.2 KB / 1

▾ DAG Visualization



▸ Show Additional Metrics
▸ Event Timeline

HelloSpark.py × utils.py ×

```python
from pyspark import SparkConf


def get_spark_app_config():
    spark_conf = SparkConf()
    config = configparser.ConfigParser()
    config.read("spark.conf")

    for (key, val) in config.items("SPARK_APP_CONFIGS"):
        spark_conf.set(key, val)
    return spark_conf


def load_survey_df(spark, data_file):
    return spark.read \
        .option("header", "true") \
        .option("inferSchema", "true") \
        .csv(data_file)


def count_by_country(survey_df):
    return survey_df \
        .where("Age < 40") \
        .select("Age", "Gender", "Country", "state") \
        .groupBy("Country") \
        .count()
```

Hello Spark - Spark Jobs

localhost:4040/jobs/

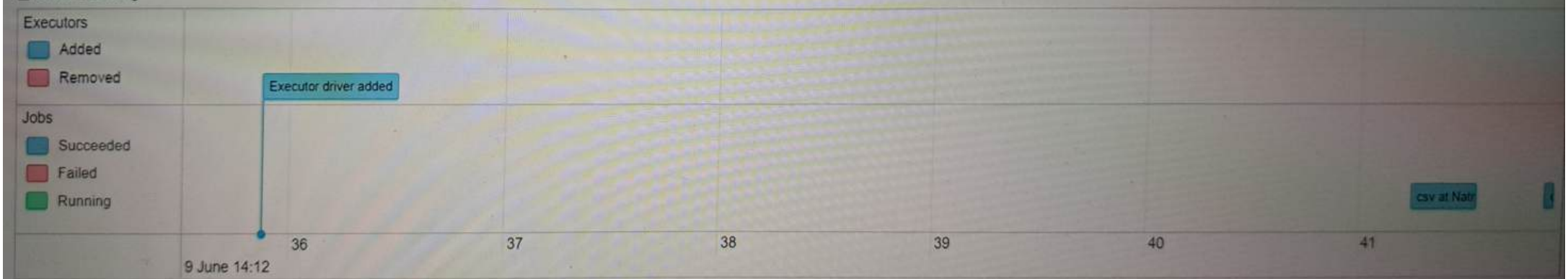
Spark 2.4.5 | Jobs | Stages | Storage | Environment | Executors | SQL | Hello Spark application

Spark Jobs (?)

User: prashant
Total Uptime: 12 s
Scheduling Mode: FIFO
Completed Jobs: 2

▾ Event Timeline

☐ Enable zooming



▾ Completed Jobs (2)

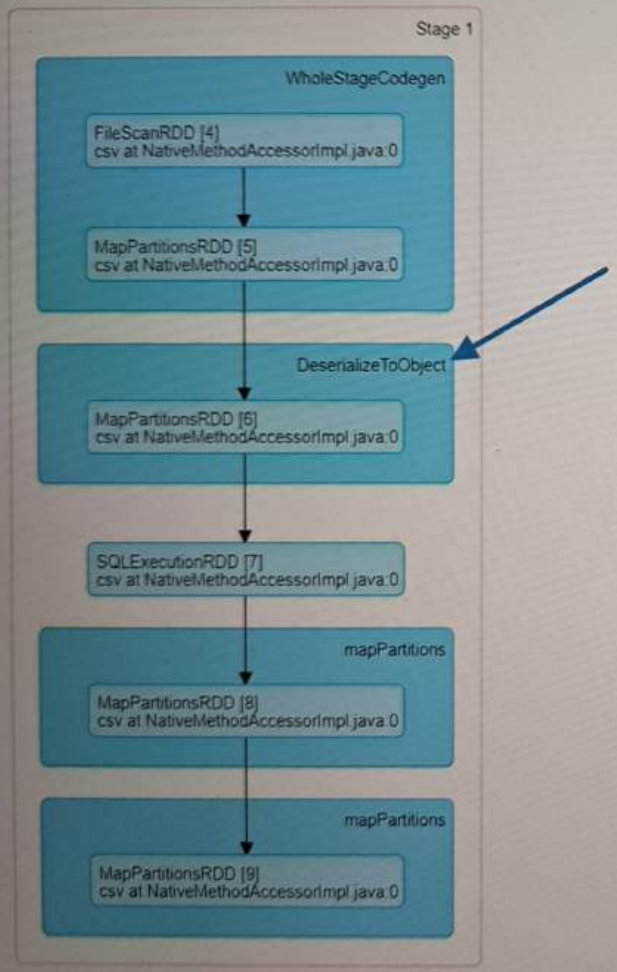
Job Id ▾	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
1	csv at NativeMethodAccessorImpl.java:0 csv at NativeMethodAccessorImpl.java:0	2020/06/09 14:12:41	47 ms	1/1	1/1
0	csv at NativeMethodAccessorImpl.java:0 csv at NativeMethodAccessorImpl.java:0	2020/06/09 14:12:41	0.3 s	1/1	1/1

Hello Spark - Details for Stage 1

localhost:4040/stages/stage/?id=1&attempt=0

Input Size / Records: 2.2 KB / 10

DAG Visualization



Show Additional Metrics

Spark Execution Plan

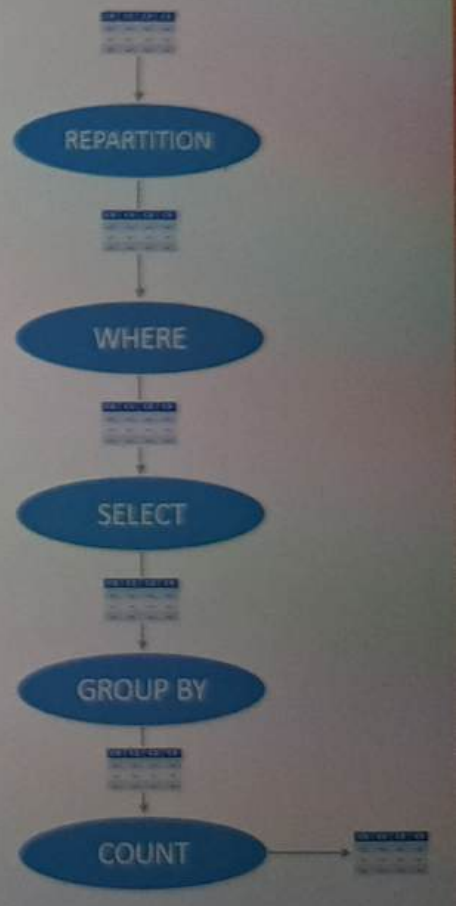
```
survey_df = spark.read \
    .option("header", "true") \
    .option("inferSchema", "true") \
    .csv(sys.argv[1])
```

Job 0

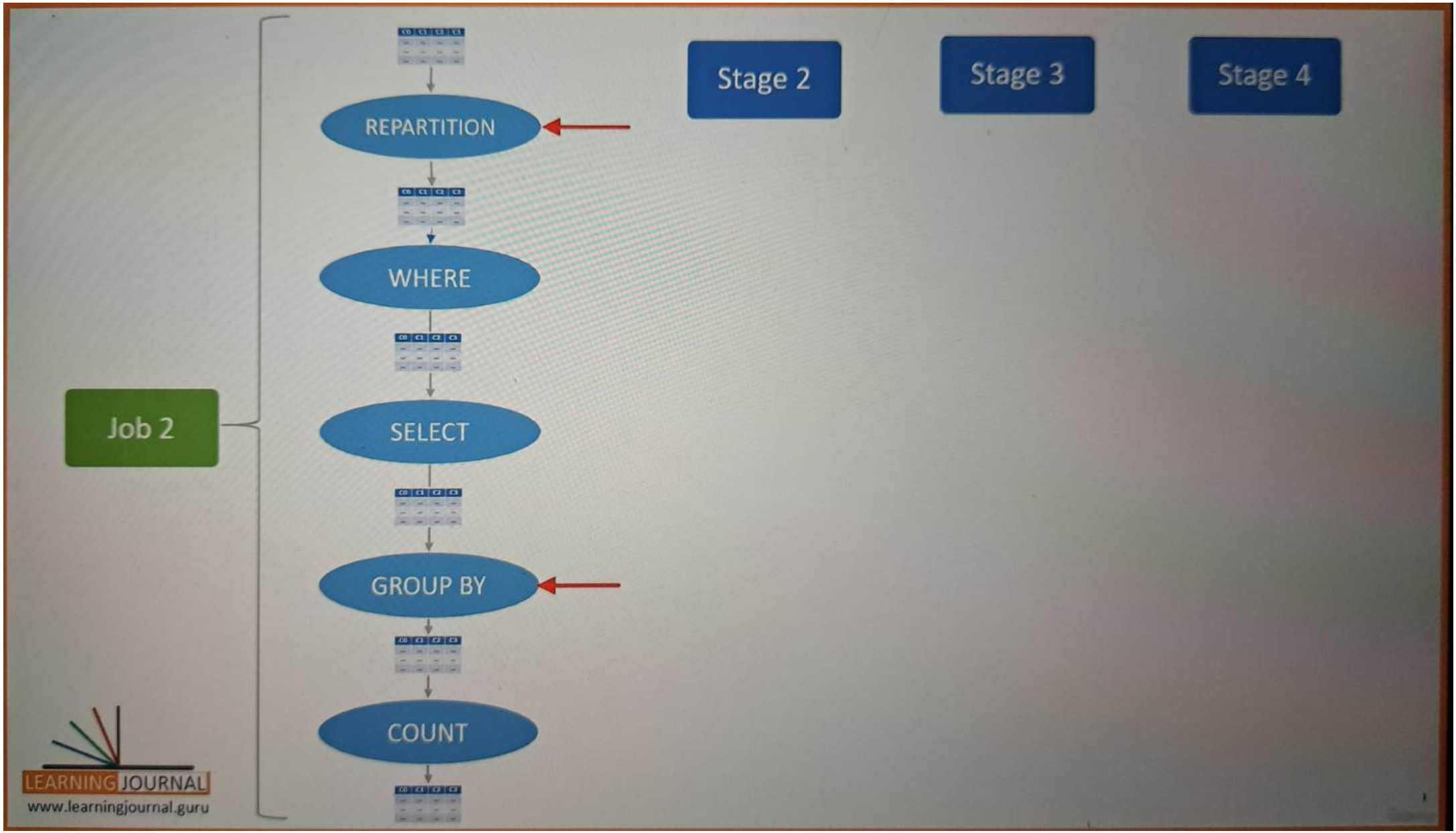
Job 1

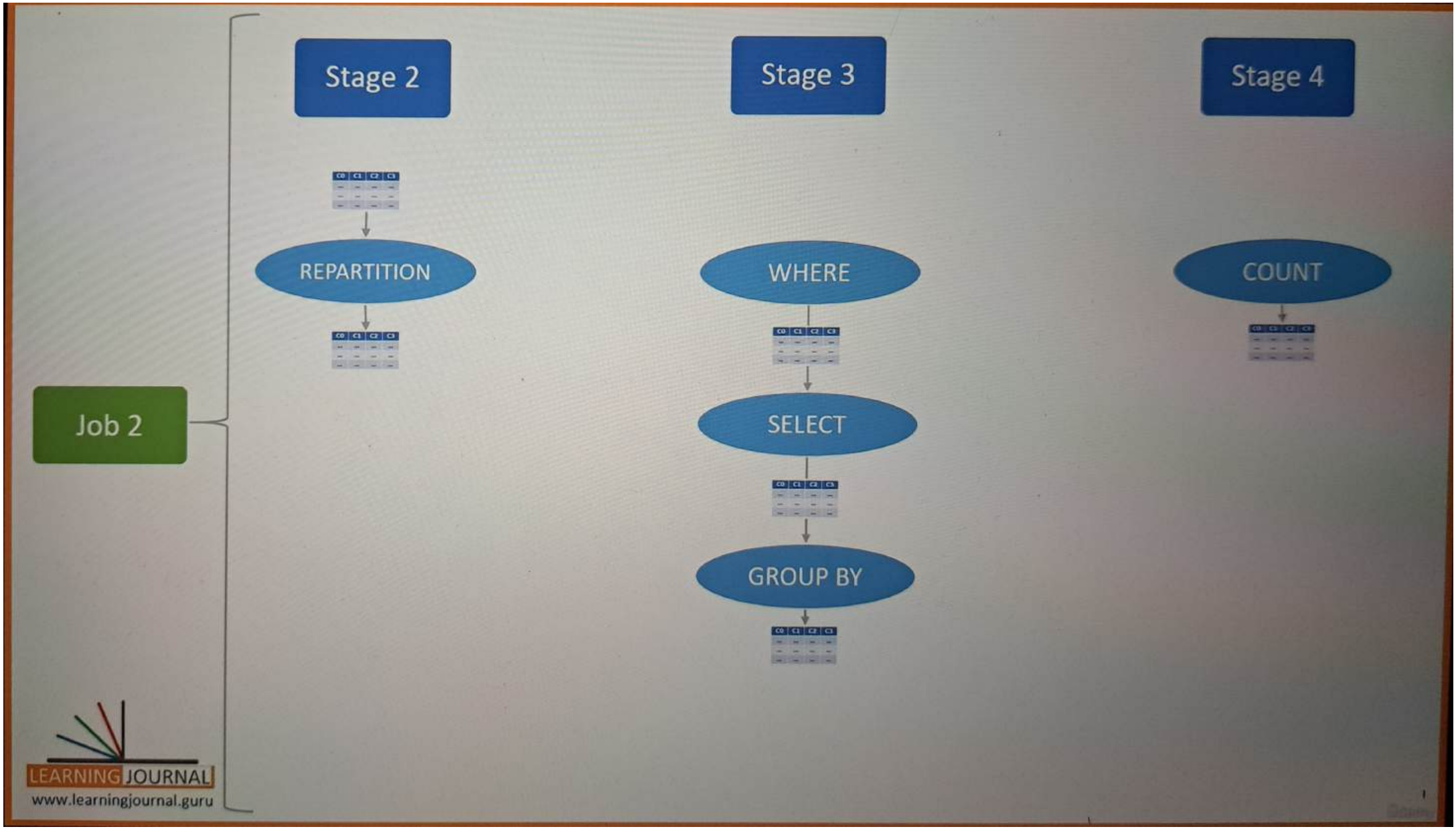
```
partitioned_survey_df = survey_raw_df.repartition(2)
count_df = survey_df.filter("Age < 40") \
    .select("Age", "Gender", "Country", "state") \
    .groupBy("Country") \
    .count()
logger.info(count_df.collect())
```

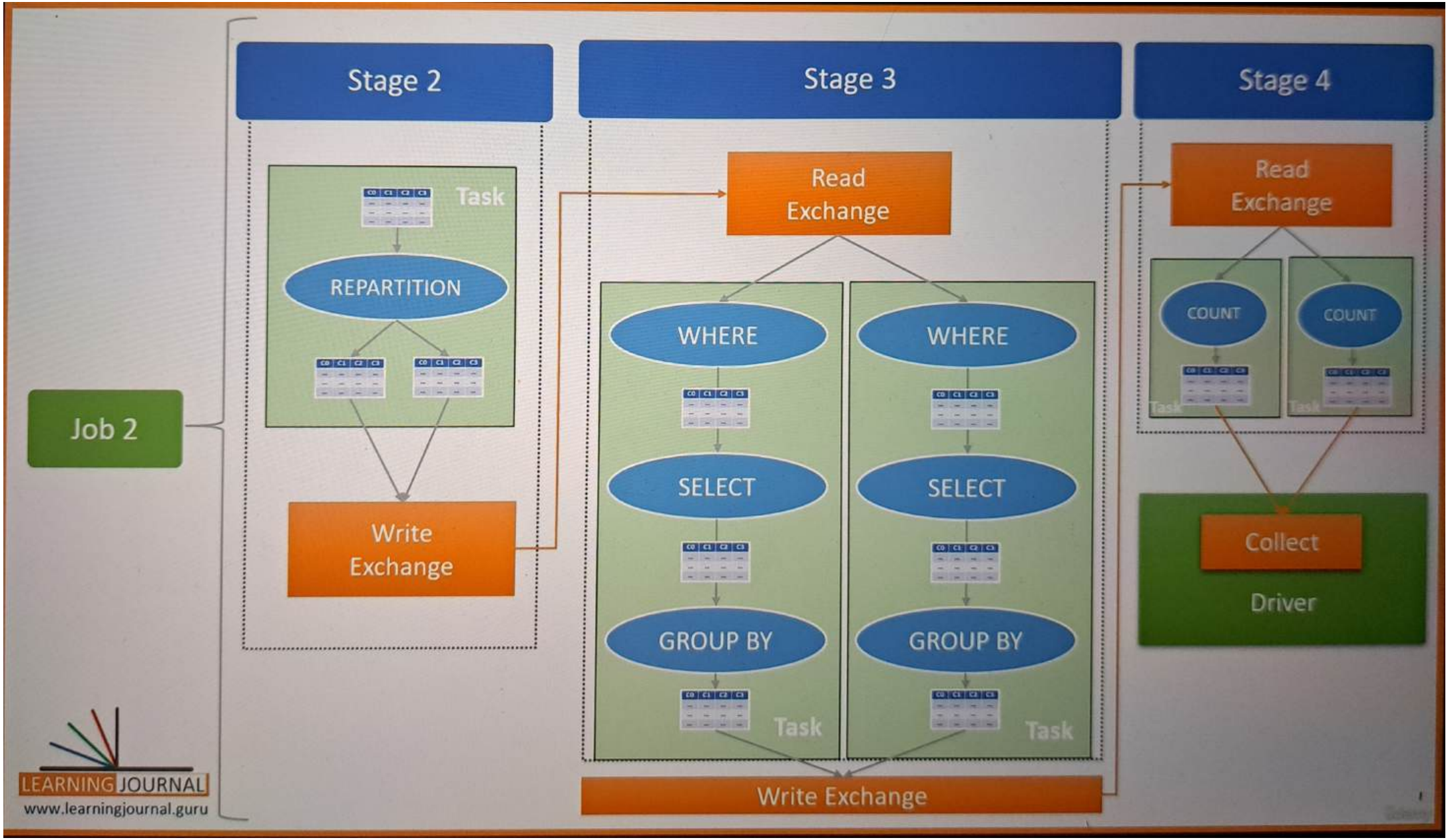
Job 2



LEARNING JOURNAL
www.learningjournal.guru


Stage 2
Stage 3
Stage 4
REPARTITION
WHERE
SELECT
GROUP BY
COUNT
Job 2
LEARNING JOURNAL
www.learningjournal.guru


Job 2
Stage 2
REPARTITION
Stage 3
WHERE
SELECT
GROUP BY
Stage 4
COUNT
LEARNING JOURNAL
www.learningjournal.guru


Job 2
Stage 2
Task
REPARTITION
Write Exchange
Stage 3
Read Exchange
WHERE
SELECT
GROUP BY
Task
WHERE
SELECT
GROUP BY
Task
Write Exchange
Stage 4
Read Exchange
COUNT
COUNT
Task
Task
Collect
Driver
LEARNING JOURNAL
www.learningjournal.guru

Hello Spark - Spark Jobs

localhost:4040/jobs/

Spark 2.4.5 | Jobs | Stages | Storage | Environment | Executors | SQL | Hello Spark application UI

Spark Jobs (?)

User: prashant
Total Uptime: 15 s
Scheduling Mode: FIFO
Completed Jobs: 3

▸ Event Timeline

▾ Completed Jobs (3)

Job Id ▾	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
2	collect at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py:27 collect at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py:27	2020/06/09 14:21:28	0.6 s	3/3	5/5
1	csv at NativeMethodAccessorImpl.java:0 csv at NativeMethodAccessorImpl.java:0	2020/06/09 14:21:27	63 ms	1/1	1/1
0	csv at NativeMethodAccessorImpl.java:0 csv at NativeMethodAccessorImpl.java:0	2020/06/09 14:21:26	0.4 s	1/1	1/1

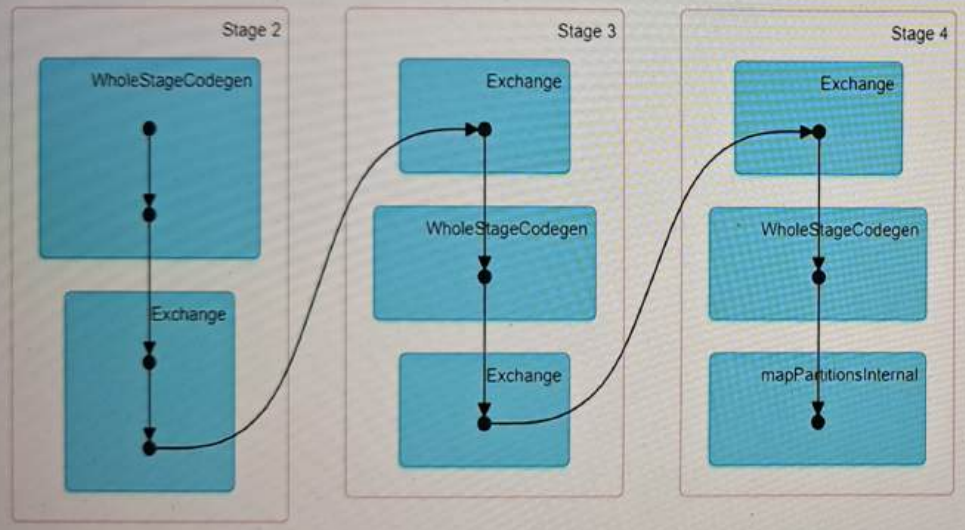
Hello Spark - Details for Job 2

localhost:4040/jobs/job/?id=2

Details for Job 2

Status: SUCCEEDED
Completed Stages: 3

- Event Timeline
- DAG Visualization



Completed Stages (3)

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
4	collect at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py:27 +details	2020/06/09 14:21:29	16 ms	2/2			229.0 B	
3	collect at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py:27 +details	2020/06/09 14:21:28	0.2 s	2/2			232.0 B	229.0 B
2	collect at C:/Users/prashant/Documents/GitHub/Spark-Programming-In-Python/HelloSpark/HelloSpark.py:27 +details	2020/06/09 14:21:28	0.3 s	1/1	2.2 KB			232.0 B